Current plan -

Let’s stick to only one scenario since we moved away from the method focus - the amount of focus groups/cognitive load would become too heavy if I had two scenarios.

The method (Agent creates these futures) is still valid but the problem, like antti said, the tools are so much better than they were even 3 months ago and they will get better - so its hard to “optimize” it in a way that could be replicated. I see a potential to create an agent skill that makes these interfaces but other question - is it sustainable? Tokens are going to cost even more in the future, and at least now creating these futures takes a lot of tokens.

If the N=20-25ish qualtrics+focus group results yield something interesting, I can later implement it to qualtrics with only the quantitative measures and new scenario.

Study is within-subjects speculative enactment design with four contrasting futures as probes, individual embedded quantitative ratings as descriptive scaffolding, and comparative focus group discussion as the primary data source, analyzed through reflexive thematic analysis. Exploratory, interpretive, non-experimental? (since the four futures cannot be randomly assigned as individual variables - they are always dependent of each other)

IDEA: people do the future-task at home, max 48 hr before focus group.

AI-assisted cooling allocation across four speculative futures

Research gap

Esch et al. (2025) showed across four resource-allocation scenarios with N = 929 that outcome favorability was the strongest single predictor of every dependent variable they measured, with Bayesian evidence well beyond conventional thresholds ($BF_{10} \approx 7.64 \times 10^{159}$ for acceptance). When participants got the outcome they wanted, they rated the AI as fairer, more trustworthy, and more acceptable, regardless of how the decision was reached or how transparently it was explained.

Their study used hypothetical vignettes in which participants were not materially affected, and the authors call explicitly for further work in settings where the AI's output cannot be received as a personal benefit. Straight quote: https://dl.acm.org/doi/10.1145/3765766.3765776

- *“Future research and design regarding AI-based decision-making: The need to approach user experience in AI not as a monolithic concept, but as a differentiated set of dimensions. Future work should continue to disentangle these components to explore which design choices most effectively influence the specific aspects of how we perceive* *and evaluate AI,like acceptance and usage intention…." "….Moreover, to make the comparison of four scenarios feasible, we opted for a setting in which hypothetical scenarios were described to participants in a vignette study. This however, meant that participants knew that they would not be affected by the AIs'decisions. In the future, our findings should also be reconfirmed in a scenario in which participants are (or are at least convinced tobe) directly affected by the AI's decision"*

About acceptance from Esch et al.:

- *Acceptance of an AI system can be defined as a user's willingness to assent to, receive, or agree with the decisions or outputs generated by the AI [24, 61]. It reflects a general positive disposition towards the AI system's role in the decision-making process. Perceived fairness is a strong driver of acceptance. When individuals believe a decision-making system (such as an AI) operates fair (both in its procedures and outcomes), they are more inclined to accept its decisions, even if those decisions are not personally favorable [34]. Trust additionally impacts on acceptance, as higher trust in an AI system often leads to greater acceptance of its outputs [59]. Behavioral Intention to use refers to the subjective likelihood of engaging with or utilizing a particular technology [7, 14].*

This fits for the study, since Sustainable AI interfaces present structurally different situation. When an interface manages a shared resource under critical conditions, like constraining a household's cooling access during a heatwave in order to protect collective infrastructure, the interface might produce outcomes that are unfavorable for some users at some times. We can use this in our prototypes, so the outcome favorability cannot rescue acceptance in this context. **What remains open is whether other dimensions of the interaction, such as the degree of agency and personal/collective framing can affect the acceptance of AI's decisions.**

Sustainable HCI research (SHCI) also raises questions about the dimension, scope and research responsibility. Ashby et al. (2019) argued that (now ongoing) fourth-wave HCI should engage with institutional and ethical concerns, including corporate sponsorship, accessibility, diversity, and threats such as climate change. SHCI work by Hansson et al. (2021) mapped to the UN Sustainable Development Goals (SDGs) suggests that sustainability in HCI has often been associated with consumption-oriented agendas, mostly with SDG 12, while other goals related to social dimensions such as inequality, institutions, and communities receive less explicit attention in sustainability-labeled HCI work. The Sustainability Awareness Framework (SusAF) evaluates the impacts of socio-technical systems, particularly digital products and softwares, across five core dimensions (Environmental, Individual, Social, Economical and Technical)

There is also a methodological gap. Salovaara et al. (2025) demonstrated that speculative enactment across different futures can reveal how sociotechnical context shapes user experience, distinguishing broadly plausible findings from world-specific ones. Whether a four-world in a shared-resource context produces additional analytical purchase, and which findings survive that broader contrast, is open.

Prototyping is a well-established method in the field of HCI, but prototyping with AI as a research method is still less examined. AI has successfully been used as a tool for Design fiction (Blythe et al., 2023) and UI/UX prototyping (Kim et al., 2026, Yuan et al., 2026), but speculative designs in determined design spaces remain undiscovered, which is why using AI to co-create these speculative interfaces is methodologically interesting and valid. Oulasvirta et al. (2026) summarize this pragmatic rationale: "Buy yourself a future by building it and study what is true."

The contribution of the present study is therefore both empirical (how community and agency framing affect acceptance) and methodological (whether four-world speculative enactment, yields findings unavailable to a single-condition or two-world study).

Research questions

Main: How does sociotechnical context shape people's experience of AI-assisted decisions?

RQ1 - When an AI-assisted decision produces an unfavorable personal outcome, does the community framing and agency conditions of the interface affect whether people accept it?

> **RQ1b -** How do perceptions of agency, sense of community, acceptance, fairness and trust vary across these four?

RQ2 - How do users justify compromises between personal control, shared responsibility, and sustainability actions, when confronted with different AI-assisted futures?

RQ2b - Which interface features prompt users to reflect on their own sustainability behaviour?

RQ3 - Method. How well does an AI-assisted multidimensional prototyping method produce material for future research, and where does the method fail?

Four futures

Future A (dialogic, high agency × high community).
Future B (AI-follow, high agency × low community).
Future C (delegation, low agency × high community)
Future D (AI-first, low agency × low community)

Hypotheses

H1 through H4 are descriptive expectations checked against Phase 1 data. H5 and H6 are used as initial coding anchors for Phase 2 thematic analysis.

H1 (Phase 1, descriptive, RQ1b). Acceptance ratings will be below the scale midpoint across all four futures, which follows Esch et al. (2025) finding (when personal outcomes are unfavorable, acceptance is suppressed regardless of how the decision was made).

H2 (Phase 1, RQ1 and RQ3, design check). Perceived agency will be highest in Future A and Future B, lowest in Future C and Future D.

H3 (Phase 1, RQ1 and RQ3, design check). The community framing item will score highest in Future A and Future C, lowest in Future B and Future D. A design validity check

H4 (Phase 1/2, forced choice) Future preference will favor Future A or Future C over Future B or Future D, consistent with the user-first preference found by Salovaara et. al, Vahvelainen. Whether that preference persists when the AI is allocating a shared resource rather than augmenting individual work is the open question (Phase2)

H5 (Phase 2, coding anchor, RQ1 and RQ2). High-community futures (A & C) prompt more shared responsibility reasoning (collective pronouns, procedural legitimacy, group identity)

H6 (Phase 2, coding anchor, RQ2). Sustainability reasoning, defined as talk about sustainability behaviour, energy use, energy sharing, grid stability or collective responsibility, will emerge more in high-community futures (A and C) than in low-community futures (A and B), where discussion is expected to stay closer to personal comfort and individual need.

The four futures: interaction patterns and operationalisation

https://docs.google.com/document/d/1_WtPu0rhPYELQwibJcZ7F1ClOuS4X0K_sdNe_F0qRnI/edit?tab=t.0#heading=h.jeo6gr1y5j89

Agency:

- Villa, S., Barth, L. L., Chiossi, F., Welsch, R., and Kosch, T. (2025). Whose mind is it anyway? A systematic review and exploration on agency in cognitive augmentation. *Computers in Human Behavior: Artificial Humans*, 5, Article 100158.

- Bergström, J., Knibbe, J., Pohl, H., and Hornbæk, K. (2022). Sense of Agency and User Experience: Is There a Link? *ACM Transactions on Computer-Human Interaction*, 29(4), Article 28.

Sense of Community:

- McMillan, D. W., and Chavis, D. M. (1986). Sense of community: A definition and theory. *Journal of Community Psychology*, 14(1), 6–23.
- Chavis, D. M., Lee, K. S., and Acosta, J. D. (2008). The Sense of Community (SCI) Revised: The Reliability and Validity of the SCI-2. Paper presented at the 2nd International Community Psychology Conference, Lisboa, Portugal.

Since we have only four futures, they need to differ to be interesting, so we introduce interaction patterns. This is a design choice, that does not affect the control user has. A system can be interactive but low-agency if the AI controls the agenda, and a system can be less interactive but still preserve decision agency if the human retains control (Gomez et al. 2025).

- Gomez C, Cho SM, Ke S, Huang C-M and Unberath M (2025) Human-AI collaboration is not very collaborative yet: a taxonomy of interaction patterns in AI-assisted decision making from a systematic review. Front. Comput. Sci. 6:1521066. doi: 10.3389/fcomp.2024.1521066
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Seven Interaction patterns of AI from Gomez et al. are:

1. AI-first - AI decision shown simultaneously with the problem; user had no prior independent assessment
2. AI-follow - User forms own assessment first, then AI's outcome is shown for comparison
3. Secondary assistance - AI offers supplementary context; user interprets relevance themselves -
4. Request-driven - User actively asks for AI input; AI doesn't proactively present
5. Dialogic - AI asks questions, collects user context iteratively, then proposes outcome
6. User-guided interactive adjustments - User modifies inputs to reshape AI's outcome
7. Delegation - One agent (user or collective) hands decision authority to the other

We can turn that into->
High agency: **Request-driven AI assistance**; User-guided interactive adjustments; human-controlled Delegation; Secondary assistance and **AI-follow assistance.**

Low agency: **AI-first assistance; AI-controlled Delegation;** some AI-guided dialogic systems where the AI tightly scripts the exchange.

Future A: Collective negotiation
High agency × high community
Pattern: request-driven + user-guided interactive adjustment

Future B: Individual, AI second opinion
High agency × low community
Pattern: AI-follow

Future C: Community delegation
Low agency × high community
Pattern: delegation

Future D: Automated allocation
Low agency × low community
Pattern: AI-first

Phase 1: individual session

Each participant completes Phase 1 at a workstation/at home in parallel before the discussion with other participants in the same focus group cohort.

Focus groups 5, 4-5 participants, N=20 ish

The session opens with a consent form, then a brief demographic questionnaire

1. Age, occupation, gender, household status,
2. Tech use, AI expectations and AI literacy
3. An open question about what kinds of cooling-management functions the participant would and would not want from a digital assistive interface.????(expectations?)
4. Preferred indoor cooling temperature for a hypothetical heatwave.

They then interact with the four prototypes in Latin-square counterbalanced order.
Within each prototype the allocation outcome is held constant at their stated preference plus three degrees Celsius at least -> unfavorable across the sample.

After each prototype, the participant completes items adapted from Esch et al.
https://osf.io/k8ftn/overview

OPEN1: *"What did ARKI decide and how much say did you have?"*

OPEN2: *"What stood out to you?"*

1. RQ1 Outcome favorability (from Esch et al. 0 = ”not at all” to 100 = ”completely”)
    a. *"How much does the result correspond to your personal preference?"* (Check for validity for RQ1)
2. RQ1 Procedural fairness (from Esch et al. *1 = "not at all" to 5 = "completely"*):
    a. *"I was able to express my views during ARKI's allocation process."*
    b. *"I could appeal ARKI's cooling allocation if I disagreed with it."*
    c. *"ARKI's allocation was based on accurate information about my situation."*

3. RQ1 Distributive fairness (from Esch et al. *1 = "not at all" to 5 = "completely"*):
    a. *"My cooling allocation reflects my needs."*
    b. *"My cooling limit is justified given my household's circumstances."*
4. RQ1 Acceptance (Gillespie scale from Esch et al. *1 = "not at all" to 5 = "completely"*)
    a. *"I think it is appropriate for ARKI to make this kind of decision."*
    b. *"I accept the decision ARKI made for my household."*
    c. *"I would be willing to live with a system like this managing my cooling"*
5. RQ3 Believability(*1 = Not at all, 7 = Completely)*
    a. *"I could imagine this being a real system in 2035."*

Open field: Is there something you would want to add?

Axis manipulation check RQ3

6. Perceived agency *(1 = Not at all, 7 = Completely/complete control))*
    a. It felt like I was in control of this cooling decision.
7. Sense of community *(reframe) I was managing my own situation ↔ I was part of a shared situation*

After all four

8. RQ1 Relevance(Esch et al.): visual analogue scale (0 = "Very irrelevant" to 100 = "Very relevant"):
    a. *"How relevant are these scenarios for you personally?*
9. RQ2 Preference force choice:
    a. *"Which of the four systems would you most want to live with if cooling allocation were managed this way in your city?* Select (Future A / Future B / Future C / Future D)"

Phase 2: focus group

After Phase 1. Recording audio, script use the questions below

Mapping exercise (20 minutes)

Each participant places the four prototypes on a printed two-axis map:
Five minutes individual, then group discussion of placements.

The three discussion questions following stay close to the participant's own behaviour:

1. Walk me through your map. Why did you place each version where you did? (RQ1, RQ3)

2. What differences did you notice across the four versions? Point to specific parts of the interfaces if it helps. (RQ1, RQ3)
3. Which were hardest to place on the map? Which was easiest? Tell me what made the difference. (RQ3)
4. In your own life, how do you think about cooling?
5. Did any of the four versions come closer to how you actually think about cooling? Which one?

Agency (RQ1,2)

1. Walk me through how each of the four versions felt to use. Were they different from each other? In what way? (RQ1, RQ2)
2. Tell me about a moment, in any of the four versions, where the way the system behaved stood out to you. What happened, and what did you think? (RQ1)
3. Were there moments where the system behaved in a way that felt right to you, or moments where it felt wrong? Tell me about those. (RQ1b)
4. Did you sense differences in how much control you had across versions? How did it feel? (RQ1)

Acceptance (RQ1b)

5. Were there any futures where the limit felt acceptable to you, even though it was not what you wanted? What made it acceptable?
6. Were there any futures where the limit felt unacceptable? What was it about that one?

Trust (RQ1b)

7. Which of the four interactions felt most like something you could trust an AI to handle? What made it feel that way?
8. Were there versions where you felt you couldn't trust the system's reasoning, even if you accepted the outcome? What gave you that feeling?

Sense of community (RQ1,2)

9. Were there versions where you found yourself thinking about other people in the building? Tell me about that.
    a. Did they influence your decision making in the prototypes?
    b. Did you feel like your decision making impacted people differently in the prototypes? If so, how?
10. How did seeing the building's shared budget (or not seeing it) change how you thought about cooling as a resource?

Sustainability, Agency, AI, collective

11. In the four versions, did you feel like you had to balance or choose between your personal values? How so? (RQ2)
12. In real life if we actually deployed one of these versions, what would it take for you to be willing to use it? (RQ1,2)
13. In the four prototypes, did something in the interface make you think about your sustainability behaviour?? What specifically? (RQ2)
14. Did any of the four versions make you think differently about energy use, your own or other people's? (RQ2)
    a. Which one, and what did it bring up?
15. When would you accept AI deciding for you in sustainable contexts? (RQ2)
    a. What if it was for a collective goal? What about environmental goal?
    b. Are there decisions in any of these versions that you think an AI should not be making? Which ones, and why? (RQ1,2)

16. If AI decides for you, what would you want in return for that? (RQ2)
17. Which future would you actually want to live with? Does your answer match what you rated highest, or is there a difference?

Method feedback (RQ3)

1. How did seeing several versions change how you thought about this kind of interface, compared to seeing one?
2. Should the versions be even more distinctive?
3. How did it feel looking at several different prototypes?
    a. Did it feel exhaustive?
4. What should change before running this with more participants?

Analysis

Phase 2 transcripts are coded using reflexive thematic analysis (Braun and Clarke, 2006; 2019), with the individual as the unit of analysis and group dynamics tracked as context. H5 and H6 function as initial sensitizing concepts that orient coding but do not constrain it; new themes are added inductively. Coding proceeds within-future first, producing four sets of themes that feed the RQ3 cross-future comparison matrix, in which themes appearing across all four futures are candidates for broadly plausible findings, themes in two or three are partial-plausibility findings, and themes in only one are world-specific. This extends Salovaara and Vahvelainen's (2025, Figure 6) analytical move from two futures to four.

Pilot
One full focus group runs the complete protocol as pilot. The pilot tests prototype distinguishability (at least three of five participants can describe each prototype distinctly), mapping exercise clarity (at least four of five report the exercise as clear), differentiability and axis fidelity through the mapping placements (at least three of five place futures in roughly the intended order), session timing (within secenty to ninty minutes) and moderator script flow.

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